

# Audio Moment Retrieval from Long Audio based on Multi-Stage Boundary Refinement and Candidate Re-Ranking

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**Abstract**—Audio moment retrieval consists of localizing the temporal segment, or segments, in a long audio recording that correspond to a natural-language query. Unlike conventional language-based audio retrieval, which typically operates at the clip level, this task requires open-vocabulary audio-text matching together with accurate temporal localization in untrimmed recordings. We propose a coarse-to-fine framework for audio moment retrieval that combines global proposal generation, local boundary refinement, and candidate-level reranking. The first stage applies a UVCOM-style temporal localization model to the full audio recording to obtain a ranked set of candidate moments. The second stage refines the most promising candidates on local crops using higher-resolution audio features, improving boundary accuracy. The final stage reranks the refined candidates using temporal descriptors, localization confidence, audio-text similarity, and boundary-context information. We additionally apply simple postprocessing steps, including timestamp rounding, minimum-duration enforcement, duplicate removal, and boundary clipping. Submitted to DCASE 2026 Task 6, our team achieved fifth place in the challenge, with an  $R1@0.7$  of 40.01 for the best single system and 42.09 for the ensemble, however in the alternative metric  $R1@0.5$  our team ranked 2nd in the evaluation with 64.41. These results indicate that decomposing audio moment retrieval into coarse localization, local refinement, and reranking is an effective strategy for language-based temporal localization in long audio recordings.

**Index Terms**—Audio moment retrieval, language-based audio retrieval, text-to-audio retrieval, long audio understanding, temporal localization, cross-modal retrieval, coarse-to-fine localization, DETR-based retrieval

## 1. INTRODUCTION

Language-based audio retrieval aims to search audio content using free-form natural-language queries. While clip-level audio-text retrieval determines whether a recording is relevant, many applications require finer temporal localization: the system must identify when the queried event occurs within a long, untrimmed signal. This is useful for locating highlights in broadcasts, finding events in surveillance recordings, or navigating multimedia archives.

Audio moment retrieval (AMR) addresses this setting by taking a long recording and a natural-language query as input and returning one or more temporal windows matching the described sound event or scene. Unlike sound event detection, AMR is not limited to a predefined taxonomy; it requires open-vocabulary matching between complex textual descriptions and localized acoustic content. The system must therefore jointly model audio-text semantic alignment and temporal structure over long sequences.

DCASE 2026 Challenge Task 6 focuses on AMR from long audio recordings and is closely related to Clotho-Moment [1] and CASTELLA [2]. Clotho-Moment introduced a simulated AMR framework and showed the advantage of DETR-style [3] temporal localization over sliding-window clip retrieval, while CASTELLA extends the task to real-world one-to-five-minute recordings with human-annotated captions and boundaries. These benchmarks highlight key difficulties: target moments can be short, repeated, overlapping, or embedded in complex backgrounds. Pretrained text-to-audio encoders

provide useful semantic representations, but they are typically trained for complete clip-caption matching rather than accurate boundary prediction, so they must be adapted to a temporal localization framework.

Our submission uses a coarse-to-fine strategy. A global UVCOM-style [4] model first identifies promising temporal regions in the full recording. A second UVCOM-style refinement model then operates on local high-resolution crops around the top candidates to improve boundary precision. Finally, a lightweight candidate reranker scores the refined candidates using temporal, confidence, audio-text similarity, and boundary-context features. We submit three single systems and one ensemble system based on this pipeline.

The individual building blocks are adapted from established retrieval and DETR-style localization methods; the contribution of this work is therefore primarily *system-level*. Specifically, we study a modular decomposition of AMR into global recall, local boundary correction, and candidate selection, and quantify the contribution of each stage on the same benchmark. This positioning is complementary to approaches that seek gains mainly through larger backbones, specialized boundary-aware architectures, or substantially larger ensembles. In this workshop version, the stage-wise results are explicitly interpreted as an ablation of that design.

## 2. SYSTEM OVERVIEW

The submitted systems formulate audio moment retrieval as open-vocabulary temporal localization in long audio. Given an audio recording represented by a sequence of window-level audio embeddings and a natural-language query represented by a text embedding, the system predicts a ranked set of temporal windows. Each prediction is a segment

$$p_i = [s_i, e_i], \quad (1)$$

with an associated confidence score  $c_i$ . The reference annotation is denoted as

$$g = [s_g, e_g], \quad (2)$$

where  $s$  and  $e$  are the start and end times of the target moment.

The main pipeline is trained and applied in stages. A coarse UVCOM model first performs global localization over the full audio recording. A Fine-UVCOM model then refines the most promising coarse candidates using local high-resolution crops. A candidate reranker finally scores the refined candidates using candidate-level temporal, semantic, confidence, and boundary-context features. The final predictions are postprocessed before submission. The complete system is therefore not trained end-to-end; instead, each stage is optimized after the previous stage has generated the inputs required for the next one.

## 2.1. Temporal overlap and notation

For a predicted interval  $p = [s_p, e_p]$  and a reference interval  $g = [s_g, e_g]$ , temporal intersection over union is  $\text{IoU}(p, g) = |p \cap g| / |p \cup g|$ . When required by the UVCOM implementation, intervals are converted from start–end coordinates to normalized center–width coordinates. Generalized temporal IoU (GIoU) is defined as  $\text{GIoU}(p, g) = |I| / |U| - (|C| - |U|) / |C|$ , where  $I = p \cap g$ ,  $U = p \cup g$ , and  $C$  is the smallest interval enclosing both  $p$  and  $g$ .

## 2.2. Coarse UVCOM localization

Given a coarse audio feature sequence  $A^{\text{coarse}}$  and a text-query embedding  $z_q$ , the coarse UVCOM model [4] predicts  $N$  proposal slots,

$$P = \{(p_i, c_i)\}_{i=1}^N, \quad (3)$$

where  $p_i = [s_i, e_i]$  is the predicted temporal span and  $c_i \in (0, 1)$  is its predicted foreground probability. This follows the DETR-style set-prediction formulation [3]: a fixed set of proposals is predicted rather than exhaustively scoring all possible start–end pairs.

During training, predicted proposals are assigned to the reference moments with the Hungarian algorithm. Let  $G$  be the set of ground-truth moments for a query and let  $\sigma$  denote a one-to-one assignment between predictions and references. The selected assignment is

$$\sigma^* = \arg \min_{\sigma} \sum_i \mathcal{C}_{\text{match}}(p_i, g_{\sigma(i)}), \quad (4)$$

with

$$\begin{aligned} \mathcal{C}_{\text{match}}(p_i, g) &= \lambda_{\text{cls}} \mathcal{C}_{\text{cls}}(c_i) + \lambda_{\ell_1} \|p_i - g\|_1 \\ &\quad + \lambda_{\text{giou}} [1 - \text{GIoU}(p_i, g)]. \end{aligned} \quad (5)$$

Because every reference moment is a foreground target, the classification component of the matching cost is the negative log foreground probability,

$$\mathcal{C}_{\text{cls}}(c_i) = -\log c_i. \quad (6)$$

The coefficients  $\lambda_{\text{cls}}$ ,  $\lambda_{\ell_1}$ , and  $\lambda_{\text{giou}}$  weight the classification, span-regression, and GIoU terms, respectively. Their numerical values are given in Section 3.

After matching, the coarse model is optimized with the corresponding classification,  $\ell_1$ , and GIoU losses. Unlike the foreground-only matching cost in Eq. 6, this classification loss also supervises unmatched proposal slots as background. At inference, the  $K$  proposals with the highest confidence  $c_i$  are retained for local refinement.

## 2.3. Fine-UVCOM local refinement

The second stage improves boundary precision by applying a second UVCOM-style model to a short crop around each retained coarse proposal. For the  $k$ -th coarse proposal  $r_k = [s_k^r, e_k^r]$ , the crop interval is

$$q_k = [\max(0, s_k^r - \delta), \min(D, e_k^r + \delta)], \quad (7)$$

where  $D$  is the recording duration and  $\delta$  is the amount of temporal context; we use  $\delta = 5$  s. The reference span is shifted to crop-local coordinates, Fine-UVCOM predicts a refined local span, and the crop offset is then added back to recover global recording time.

Each crop is characterized by its proposal IoU,

$$u_k^{\text{prop}} = \text{IoU}(r_k, g), \quad (8)$$

i.e., the IoU between the coarse proposal that generated the crop and the ground-truth moment. This quantity is used only to construct refinement-stage supervision; it is not the IoU of the refined output.

Among the  $K$  crops from the same audio–query pair, the crop with the largest  $u_k^{\text{prop}}$  receives the standard UVCOM localization

loss in crop-local coordinates. Crop quality, low-overlap background rejection, and selection among the  $K$  crops are supervised by the following auxiliary losses.

More precisely, let  $a_k = \text{sigmoid}(s_k^{\text{cand}})$  be the predicted crop quality, let  $k^* = \arg \max_k u_k^{\text{prop}}$ , and let  $\mathcal{N} = \{k : u_k^{\text{prop}} \leq \theta_{\text{neg}}\}$  be the set of background crops. If  $\hat{c}_{k,m}$  denotes the foreground probability of the  $m$ -th Fine-UVCOM proposal slot in crop  $k$ , with  $M$  proposal slots per crop, the three auxiliary losses are

$$\begin{aligned} \mathcal{L}_{\text{cand}} &= -\frac{1}{K} \sum_{k=1}^K [u_k^{\text{prop}} \log a_k + (1 - u_k^{\text{prop}}) \log(1 - a_k)], \\ \mathcal{L}_{\text{neg-bg}} &= -\frac{1}{|\mathcal{N}|M} \sum_{k \in \mathcal{N}} \sum_{m=1}^M \log(1 - \hat{c}_{k,m}), \\ \mathcal{L}_{\text{topk}} &= -\log \frac{\exp(s_{k^*}^{\text{cand}})}{\sum_{j=1}^K \exp(s_j^{\text{cand}})}. \end{aligned} \quad (9)$$

The negative-background loss is set to zero when  $\mathcal{N}$  is empty.

The complete refinement objective is

$$\begin{aligned} \mathcal{L}_{\text{fine}} &= \mathcal{L}_{\text{loc}}^+ + \lambda_{\text{cand}} \mathcal{L}_{\text{cand}} + \lambda_{\text{neg}} \mathcal{L}_{\text{neg-bg}} \\ &\quad + \lambda_{\text{topk}} \mathcal{L}_{\text{topk}}, \end{aligned} \quad (10)$$

where  $\mathcal{L}_{\text{loc}}^+$  is the standard UVCOM localization loss on the span-positive crop and the remaining terms are defined in Eq. 9. The loss weights and threshold  $\theta_{\text{neg}}$  are specified in Section 3.

## 2.4. Candidate reranking

After refinement, each query has  $K$  candidate spans  $p_1, \dots, p_K$ . Detector confidence alone does not necessarily order these candidates by temporal accuracy, so a separate reranker predicts a scalar score  $r_i = f_{\theta}(x_i)$  from a feature vector  $x_i$ . The features include temporal descriptors, coarse and fine localization confidences, pooled audio–text similarity evidence, and features from the regions immediately around the candidate boundaries.

For training, the target quality of candidate  $i$  is its IoU with the reference,  $u_i = \text{IoU}(p_i, g)$ . These values are converted to a soft target distribution

$$y_i = \frac{\exp(u_i / \tau)}{\sum_{j=1}^K \exp(u_j / \tau)}, \quad (11)$$

where  $\tau$  controls how strongly the target distribution concentrates on the highest-IoU candidate. The soft-ranking loss is the cross-entropy between  $y_i$  and the softmax-normalized reranker scores.

We also use a pairwise margin-ranking loss. Let

$$\mathcal{P} = \{(i, j) : u_i - u_j \geq m_{\text{iou}}\} \quad (12)$$

be the set of candidate pairs with a sufficiently large quality difference. Candidate  $i$  is preferred to candidate  $j$ , and the pairwise loss is

$$\mathcal{L}_{\text{pair}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \max(0, m_{\text{score}} - r_i + r_j), \quad (13)$$

where  $m_{\text{score}}$  is the required score margin; the loss is set to zero when  $\mathcal{P}$  is empty. Pairs whose IoUs differ by less than  $m_{\text{iou}}$  are omitted, avoiding arbitrary ordering of nearly equivalent candidates. The final reranker objective is

$$\mathcal{L}_{\text{reranker}} = \mathcal{L}_{\text{soft}} + \lambda_{\text{pair}} \mathcal{L}_{\text{pair}}. \quad (14)$$

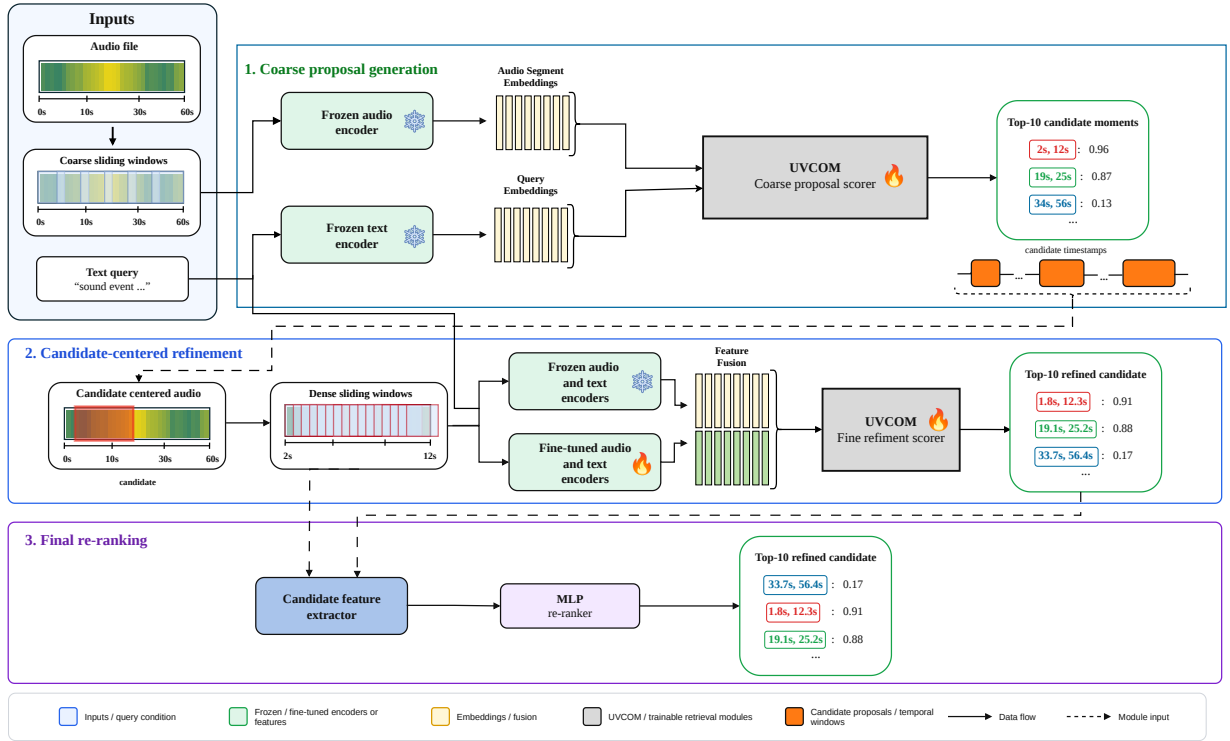


Fig. 1: Overview of the proposed audio moment retrieval pipeline. The main path is coarse localization, local refinement, candidate reranking, and postprocessing.

## 2.5. Inference-time score blending

At inference, reranker and Fine-UVCOM scores are combined as

$$\text{score}_i^{\text{final}} = r_i + \alpha d_i, \quad (15)$$

where  $d_i$  is the Fine-UVCOM detector score and  $\alpha$  is tuned on development data. This blending is used only for inference and is not part of the reranker training objective.

## 2.6. Stage-wise optimization

The pipeline is optimized sequentially rather than end-to-end. First, the coarse UVCOM model is trained on full recordings. It is then frozen and used to generate the top- $K$  proposals that define the local crops for Fine-UVCOM. After the refinement model is trained, both localization models are frozen and used to generate refined candidate lists for reranker training. The blending weight  $\alpha$  is finally selected to maximize R1@0.7 on development data.

Inference follows the same sequence: global localization, local refinement, reranking, score blending, and postprocessing. This alignment between training and inference is important because Fine-UVCOM sees crops produced by the actual coarse model and the reranker sees candidate lists produced by the actual refinement model. The main limitation is error propagation: a target moment absent from the coarse top- $K$  set cannot be recovered by later stages. The ensemble partly mitigates this failure mode by pooling proposals from models using different audio representations.

## 3. EXPERIMENTAL SETUP

Each audio-text model consists of an audio encoder, either EAT or BEATs, paired with a RoBERTa text encoder [5]–[7]. The two encoders produce modality-specific vector representations that are projected into a shared embedding space and aligned using a bidirectional

contrastive objective. We trained these EAT-RoBERTa and BEATs-RoBERTa models on Clotho [8], TACOS weak [9], AudioCaps [10], and WavCaps [11]. We followed the curriculum in [12] and added the cross-encoder estimated-correspondence stage from [13], using an auxiliary PaSST-RoBERTa model [7], [14]. Audio was divided into non-overlapping 10-second segments, independently encoded and mean-pooled. Training lasted 20 epochs with batch sizes 24 (EAT) and 20 (BEATs), temperature 0.05, AdamW weight decay  $10^{-4}$ , one warmup epoch, and a cosine learning-rate schedule from  $2 \times 10^{-5}$  to  $10^{-7}$ . These CASTELLA-unadapted embeddings were used for all coarse-stage models. The encoders were then fine-tuned for the fine stage with temporal supervision from TACOS strong and CASTELLA. Audio was represented with 2-second windows and a 0.5-second hop; embeddings overlapping each annotated moment were pooled and aligned with the query using the TACOS moment-level contrastive loss [9]. Fine-UVCOM receives both representations by concatenating the CASTELLA-unadapted embeddings with the temporally fine-tuned embeddings.

For coarse localization, UVCOM was trained on full recordings from TACOS strong, Clotho-Moments, and CASTELLA using 2-second windows with a 1-second hop. We trained EAT, BEATs, and EAT+BEATs models with 10 proposal slots and retained the top  $K = 10$  candidates for refinement. All coarse models used the classification, L1, and temporal GIoU losses from Section 2, with training and Hungarian matching weights  $\lambda_{\text{cls}} = 4$ ,  $\lambda_{\ell_1} = 10$ , and  $\lambda_{\text{giou}} = 1$ . We used AdamW with learning rate  $10^{-4}$ , weight decay  $10^{-4}$ , batch size 128 for EAT and BEATs, batch size 96 for EAT+BEATs, and trained for 300 epochs.

Fine-UVCOM was trained only on CASTELLA crops generated from coarse candidates. Each crop added 5 seconds of context on both sides of the coarse proposal and was clipped to the audio duration. The refiner used 2-second windows with a 0.5-second hop and concatenated

**Table 1:** Stage-wise results of the submitted systems on the CASTELLA test set.

| Stage           | Model     | Recall1@0.5  | Recall1@0.7  | mAP (avg)    | mAP@0.5      | mAP@0.75     |
|-----------------|-----------|--------------|--------------|--------------|--------------|--------------|
| <i>Coarse</i>   | EAT       | 42.95        | 28.64        | 22.68        | 39.64        | 21.34        |
|                 | BEATs     | 47.22        | 31.33        | 23.92        | 41.56        | 22.56        |
|                 | EAT+BEATs | 46.40        | 31.55        | 23.91        | 42.19        | 22.20        |
| <i>Fine</i>     | EAT       | 47.29        | 34.45        | 25.33        | 41.61        | 24.86        |
|                 | BEATs     | 46.18        | 33.26        | 24.06        | 40.34        | 23.81        |
|                 | EAT+BEATs | 47.51        | 35.04        | 25.39        | 41.93        | 24.74        |
| <i>Reranker</i> | EAT       | 56.12        | 40.01        | 30.17        | 47.16        | 30.39        |
|                 | BEATs     | 53.30        | 36.82        | 28.00        | 45.96        | 27.41        |
|                 | EAT+BEATs | 54.27        | 37.56        | 29.66        | 47.42        | 29.53        |
| <i>Ensemble</i> | Ensemble  | <b>59.39</b> | <b>42.09</b> | <b>33.64</b> | <b>48.73</b> | <b>34.03</b> |

non-fine-tuned and fine-tuned encoder features; for EAT+BEATs this concatenates pretrained and fine-tuned EAT and BEATs features. Training used crop-local coordinates and the same localization weights as the coarse model. For each audio–query pair, the coarse proposal with the largest proposal IoU  $u_k^{\text{prop}}$  supplied the span-positive crop. Crops with no ground-truth overlap or  $u_k^{\text{prop}} \leq \theta_{\text{neg}} = 0.05$  were supervised as background. The crop-quality and top- $K$  terms used the relative proposal IoUs within the same group of  $K$  crops. The auxiliary loss weights were  $\lambda_{\text{cand}} = \lambda_{\text{neg}} = \lambda_{\text{topk}} = 1$ , and the remaining training configuration matched coarse localization.

The candidate reranker was trained using the top  $K = 10$  refined candidates for each query. Each candidate was represented with temporal features, detector confidence features, audio-text similarity features, and context/boundary features. The temporal features included start time, end time, duration, log-duration, normalized position, center time, and width. The detector features included coarse and Fine-UVCOM scores. The audio-text features included pooled audio embeddings inside the candidate, the query embedding, and cosine similarities between them. Context and boundary features were computed by pooling audio embeddings in the left context, right context, start-boundary region, and end-boundary region.

The reranker was implemented as an MLP with LayerNorm, GELU activations, dropout 0.1, and a scalar output. All rerankers used AdamW with learning rate  $1 \times 10^{-3}$  and weight decay  $1 \times 10^{-4}$ . EAT and EAT+BEATs used MLP shape [30, 128, 128, 1], while BEATs used [30, 64, 64, 1]. The objective combined the soft IoU-label ranking loss with the pairwise hinge loss in Eq. 13. The temperatures were  $\tau = 0.10$  for EAT and  $\tau = 0.05$  for BEATs and EAT+BEATs. For EAT and EAT+BEATs,  $\lambda_{\text{pair}} = 1.0$  and  $m_{\text{iou}} = 0.05$ ; for BEATs,  $\lambda_{\text{pair}} = 1.5$  and  $m_{\text{iou}} = 0.07$ . We used the standard unit score margin  $m_{\text{score}} = 1$  for all rerankers.

At inference time, each single system generated 10 coarse candidates, refined them with Fine-UVCOM, reranked the refined candidates, and applied final postprocessing. When used, score blending added the Fine-UVCOM detector score to the reranker score with  $\alpha = 0.40$  for EAT,  $\alpha = 0.30$  for BEATs, and  $\alpha = 0.50$  for EAT+BEATs.

An ensemble model was created by merging the candidate lists produced by the three single systems. Specifically, we took the top 10 candidates from EAT, BEATs, and EAT+BEATs, yielding at most  $K_{\text{ens}} = 30$  candidates per query. Within each source system, raw candidate scores were normalized with a z-score transformation and passed through a sigmoid before merging. Exact duplicate windows were removed using identical rounded start–end keys. A separate learned MLP reranker was then trained to score the merged top-30 candidate set, using MLP shape [57, 128, 128, 1] with LayerNorm on the 57-dimensional input, GELU activations, and dropout rate 0.1 after each hidden layer. The ensemble reranker used AdamW

optimization with learning rate  $1 \times 10^{-3}$  and weight decay  $1 \times 10^{-4}$ ,  $\tau = 0.05$ ,  $\lambda_{\text{pair}} = 1.5$ , and  $m_{\text{iou}} = 0.05$ . The ensemble final score was an additive blend of the ensemble reranker score and proposal score with  $\alpha = 0.10$ .

The final predictions were rounded to the nearest second, constrained to a minimum duration of one second, deduplicated, and clipped to the audio duration.

#### 4. RESULTS AND CONCLUSIONS

Table 1 reports the performance of the submitted systems on the CASTELLA test set. Since each block adds one component to the previous stage, the table can also be read as an ablation of the proposed pipeline. The coarse UVCOM stage provides the initial localization results, with BEATs and EAT+BEATs obtaining the strongest coarse R1@0.7 values. Fine-UVCOM improves all systems, reaching an R1@0.7 of 35.04 with EAT+BEATs; the gains are most visible at the stricter R1@0.7 criterion, suggesting that the local crop stage mainly improves boundary placement rather than only retrieving additional rough matches.

The reranker gives the largest single-stage gain. EAT reaches the best single-system result, R1@0.7 of 40.01 and mAP@0.75 of 30.39, indicating that the refined top- $K$  list often contains a high-quality segment that is not assigned the highest detector score. Candidate-level duration, position, audio–text similarity, and boundary-context features therefore add information that complements detector confidence. The ensemble further improves performance to R1@0.7 of 42.09, mAP of 33.64, and mAP@0.75 of 34.03 by pooling complementary proposals from the three backbones. Compared with the best performing systems in the challenge, our system does not rely on a substantially different detector architecture or a very large model pool; its main contribution is to show that local refinement and explicit candidate reranking can provide large gains on top of a UVCOM-style localization model, even using simpler and faster audio encoders such as BEATs.

Our team placed fifth in DCASE 2026 Task 6, with an R1@0.7 of 40.01 for the best single system and 42.09 for the ensemble. The ensemble’s R1@0.5 of 64.41 ranked second, suggesting reliable retrieval of the correct region but remaining scope for finer boundary estimation. Unlike higher-ranked submissions using larger audio–language backbones, specialized temporal architectures, or substantially larger ensembles [15]–[17], our approach retains a conventional UVCOM backbone and combines only three EAT- and BEATs-based systems. Thus, local refinement and localization-aware reranking provide a competitive and comparatively simple AMR solution.

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